

TECHNICAL WHITE PAPER

RAEL Technical White Paper

Version 1.0

A Reference Architecture for Relationship-Preserving Accounting Automation

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Document status

Public technical specification. Version 1.0 defines the architectural vocabulary, processing invariants, evidence payload and reference application. It does not report empirical validation.

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Executive Summary

Accounting automation is commonly designed to convert documents and transactions into coded entries. That approach can be efficient while remaining incomplete. A payment may produce the correct debit and credit yet be applied to the wrong customer, invoice, period, credit balance or tax component. The missing object of control is the relationship through which the economic event acquires accounting meaning.

This white paper specifies the Relational Accounting Execution Layer (RAEL), an AI-assisted architecture operating between source economic events and general-ledger representation. RAEL retrieves accounting objects, forms candidate relationships, structures relevant evidence, determines whether execution is institutionally permitted, performs bounded subledger actions, and verifies the resulting state before closure.

The architecture is governed by Relationship-Preserving Accounting Automation (RPAA). Routine, evidence-complete and independently verifiable transactions may proceed through Bounded Straight-Through Processing (BSTP). Ambiguous or high-risk events move to an ASLTP-informed professional-review interface. Each transition is recorded in a Governed Execution Evidence Package (GEEP), while EGL supplies the verification and closure logic that prevents command completion from being mistaken for accounting completion.

Accounts-receivable payment matching is used as the reference implementation. The same logic can extend to accounts payable, inventory movements, payroll, fixed assets and other operational domains in which an event must be matched, authorized, executed, verified and closed across changing records.

1. The Technical Problem

Current accounting systems contain strong components for document extraction, classification, anomaly detection and posting. These components typically optimize an output: an extracted field, predicted account, proposed match or journal entry. Operational accounting, however, depends upon a network of objects and states. A bank receipt must be connected to a payer, customer, obligation, invoice or credit, reporting period, evidence source and resulting balance. The journal entry is a consequence of that structure rather than a substitute for it.

This distinction follows the relational insight of the REA model, which represented resources, events, agents and their connections rather than reducing accounting data to conventional accounts (McCarthy 1982). RAEL addresses a different but complementary question: how may an AI-enabled system act upon those relationships inside existing platforms without converting a probabilistic inference into unauthorized execution?

The architecture responds to six recurring failure conditions: correct amount applied to the wrong counterparty; correct counterparty linked to the wrong invoice; correct relationship recognized in the wrong period; duplicate application of the same event; technically successful posting without sufficient source evidence; and correct ledger classification attached to an incorrect tax or operational

relationship. These failures propagate into aging, customer statements, inventory or payable balances, cutoff, tax filings, audit evidence and management decisions.

Core distinction

Confidence answers: “How strongly does the available information support this candidate relationship?” Authority answers: “May the system act on this relationship under the applicable evidence, rule, risk and delegation conditions?” The first cannot create the second.

2. Architectural Vocabulary

Version 1.0 fixes six components. They are complementary and should not be used interchangeably.

Component	Type	Technical role
RPAA	Design principle	Preserve identifiable and verifiable links from event and evidence through obligation, settlement, state and ledger consequence.
RAEL	Execution architecture	Coordinate retrieval, classification, structuring, routing, execution and verification between source systems and the general ledger.
BSTP	Execution mode	Permit routine automatic processing only inside evidence, rule, risk, authority and verification boundaries.
GEEP	Data artifact	Maintain the machine-readable evidence, decision, authority, action and observed-state record for each event.
ASLTP	Professional interface	Present unresolved relationships and conflicting evidence for structured human judgment.
EGL	Closure logic	Compare expected and observed states and prevent closure until required verification conditions are satisfied.

2.1 Relational state model

RAEL represents a transaction as a governed transition across linked objects rather than as an isolated entry. At a minimum, the model distinguishes the economic event, counterparty, obligation, evidence, settlement instrument, subledger state and ledger consequence. Each object has an identifier and state; each link has a type, source, effective period and evidentiary basis. A payment-to-invoice link, for example, is different from a payment-to-customer link that leaves an unapplied credit. Both may create a cash consequence, but they alter different obligations and produce different customer-facing and reporting states.

The state model also makes time explicit. Events may be received, proposed, authorized, executed, verified, closed, reopened or reversed. A later reversal does not erase the earlier transition; it creates a linked compensating transition. This append-oriented approach preserves history and prevents a current balance from concealing how that balance was formed. It also supports idempotency: an event identifier

already associated with a completed or active transition cannot be executed again without a defined reversal or correction path.

Relational correctness is therefore evaluated separately from arithmetic and classification correctness. The amount must be valid, the accounting category must be appropriate, the connected objects must be the intended objects, the evidence must support the connection, the effective period must be correct, and the actor or system must possess authority for the transition. RAEI treats these dimensions as distinct predicates because success in one dimension cannot compensate for failure in another.

3. RAEI Reference Architecture

RAEI is platform-independent at the architectural level and interface-specific at implementation. It may be deployed over an ERP, accounting package, warehouse system, bank portal or collection of legacy applications. It does not replace those systems. It governs the transition by which admitted events become authorized and verified changes to operational accounting state.

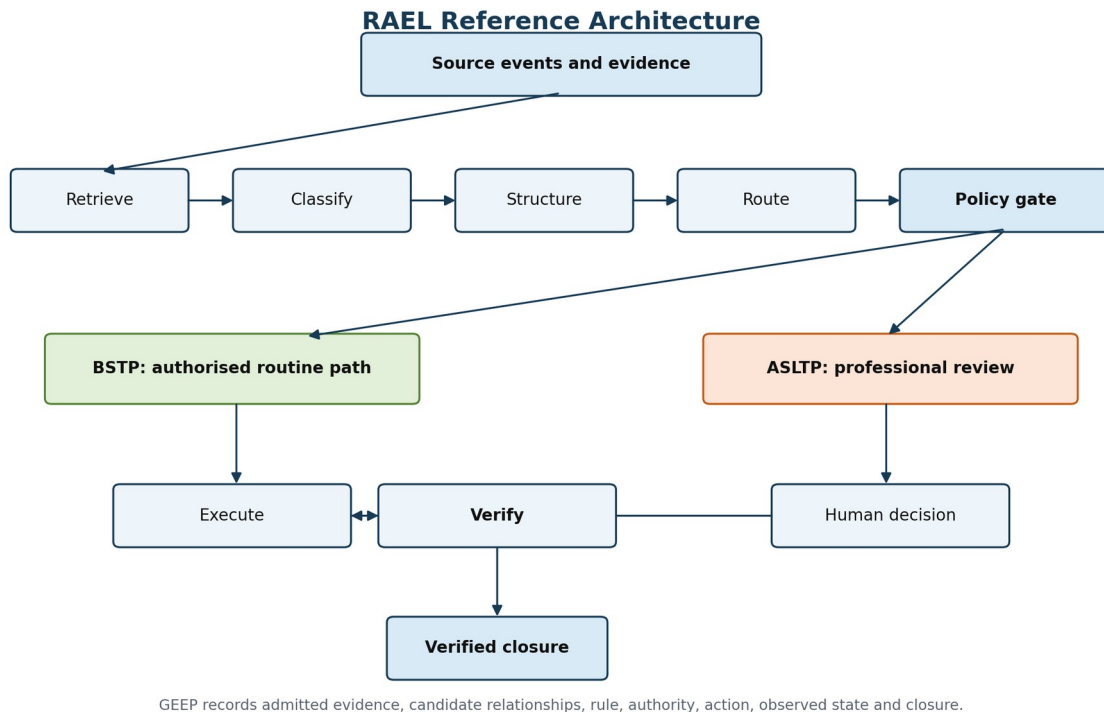


Figure 1. RAEI separates candidate formation from the policy gate and makes verified closure the terminal state.

3.1 Six-stage processing cycle

- **Retrieve.** Admit source documents, transactions, counterparties, invoices, credits, account history and contextual records from approved systems.
- **Classify.** Identify the event type, accounting objects, candidate relationships and possible exception categories.
- **Structure.** Assemble the evidence package, candidate set, inconsistencies, missing elements and proposed treatment.

- **Route.** Evaluate evidence sufficiency, rule determinacy, risk, materiality, delegated authority and post-action verifiability.
- **Execute.** Perform only the subledger action authorized for the applicable route and authority profile.
- **Verify.** Re-read the affected system state and confirm linkage, balance, status, ledger effect, audit record and closure condition.

3.2 Architectural invariants

An implementation conforms to Rael only if it preserves the following conditions:

- No execution authority arises from model confidence alone.
- No action occurs outside a defined rule, permission and risk boundary.
- No closed status arises solely from a successful command or interface response.
- Every exception identifies the relationship, evidence or authority question preventing closure.
- Every executed state transition remains reconstructable from admitted evidence to observed consequence.

4. Governed Execution Evidence Package

GEEP is the persistent data artifact connecting information formation, execution and verification. It is not merely a log of messages. Required fields function as closure predicates: where evidence, authority or observed state is absent, the event cannot be represented as fully closed. The payload may be stored as JSON, a normalized database structure, or an equivalent machine-readable object.

```
{
  "geep_version": "1.0",
  "source_event_id": "BANK-2026-004812",
  "event_type": "ar_payment",
  "candidate_objects": [
    {"type": "customer", "id": "C-1042", "score": 0.98},
    {"type": "invoice", "id": "INV-8821", "amount": 12500.00}
  ],
  "evidence": {"supporting": ["bank_reference", "remittance_advice"],
    "conflicting": [], "missing": []},
  "execution_policy": {
    "rule_version": "AR-MATCH-1.0",
    "evidence_sufficient": true,
    "rule_determinate": true,
    "risk_tier": "low",
    "delegated_authority": "BSTP-AR-01",
    "post_action_verifiable": true
  },
  "route": "BSTP",
  "proposed_action": {"apply_payment": 12500.00, "invoice_id": "INV-8821"},
  "pre_state": {"invoice_open": 12500.00},
  "observed_post_state": {"invoice_open": 0.00},
  "verification": {"status": "passed",
    "tests": ["payment_link", "invoice_balance", "customer_balance", "gl_effect"]},
  "closure_status": "closed"
}
```

Listing 1. Minimal illustrative GEEP payload. Production schemas require timestamps, actor and system identifiers, hashes or equivalent integrity controls, exception history, reversal links and jurisdiction-specific fields.

4.1 Evidence-package lifecycle and integrity

A GEEP is created when a source event is admitted, not after processing has finished. Retrieve appends source identifiers and provenance; Classify appends candidates and model metadata; Structure records the evidence matrix and missing elements; Route records the policy version, triggers and authority decision; Execute records the requested and acknowledged action; Verify records fresh observations and test results. Human review adds the question presented, evidence considered, decision, reason and reviewer authority. This sequencing allows a reviewer to reconstruct what was available at each point rather than reading a retrospective narrative.

Production implementations should protect package integrity through immutable or append-only records, trusted timestamps, stable object identifiers, versioned policies and cryptographic hashes or equivalent controls. Corrections should reference the superseded event and explain the correction basis. Access controls should separate those who configure rules, approve exceptions, operate connectors and administer logs. Retention must follow the applicable accounting, tax, privacy and audit requirements. These controls do not prove that an accounting judgment was correct; they protect the evidence needed to evaluate that judgment and the execution that followed.

5. Execution Governance

5.1 Bounded straight-through processing

BSTP is available only to low-risk, rule-conforming, evidence-complete, authorized and independently verifiable transactions. Eligibility is determined transaction by transaction. The architecture does not prescribe a universal automation percentage because evidence quality, tolerance, materiality, platform control and risk differ across entities. Automatic closure remains transparent: the input, applicable rule, authority, rationale, action, before-and-after state and verification result must remain available.

5.2 Management by exception and human control

Routine eligible cases proceed without repetitive human confirmation. Ambiguous, incomplete or high-risk cases are intercepted before execution or closure. The reviewer receives the evidence conflict and unresolved accounting question rather than a bare recommendation. Human roles vary by processing condition: human-in-the-loop for exceptional cases, human-on-the-loop for monitored routine processing, and human-in-command for rules, thresholds, permissions and system redesign (Parasuraman, Sheridan, and Wickens 2000).

Human authority therefore includes rule setting, threshold calibration, exception adjudication, sampling, audit-log review, pause, override, rollback and redesign. This placement of control avoids both unrestricted automation and the inefficiency of requiring a human click on every routine event.

6. Platform Interfaces and Resilience

Structured APIs and controlled database interfaces are the preferred channels for reading records, performing authorized actions and verifying state. They expose stable identifiers and machine-readable

responses. Computer vision, browser automation and desktop automation serve as compensating interfaces where legacy systems, bank portals or third-party applications do not provide adequate API coverage. They may also ground the system in operator-visible interface state.

Interface diversity does not change the verification requirement. Visual or browser-mediated execution is more exposed to latency, shifted elements, stale screens and interface changes. RAEL therefore requires a fresh read of the affected records after action. Where the observed relationship and balance cannot be confirmed, the event remains conditional, escalated or reversed rather than closed.

7. Reference Application: Accounts-Receiveable Payment Matching

When a payment enters the bank feed, RAEL retrieves the transaction, remittance, customer master, open invoices, unapplied credits and prior history. Classification produces candidate customers and invoices. Structuring records exact evidence, conflicts and any unresolved amount. The policy gate then determines whether the payment may proceed through BSTP.

A one-invoice payment may execute automatically where customer identity is confirmed, the amount is fully explained, the invoice relationship is determinate, no duplicate exists, delegated authority covers the action and the post-action state can be re-read. Competing customers, unexplained differences, cross-period issues, unusual credits, materiality triggers, related parties or tax-sensitive treatment move to professional review.

Execution applies the receipt to the authorized invoice. Verification then confirms the payment link, remaining invoice balance, customer balance, subledger update, general-ledger consequence and audit record. The resulting state may be closed, conditionally closed, escalated, returned for evidence, rejected or reversed. These status labels distinguish completion of a software action from resolution of the accounting relationship.

7.1 Payment-matching state machine

The reference implementation uses explicit transition guards. A newly retrieved bank item enters an unclassified state. It becomes structured only when the system has created a candidate set and evidence inventory. A BSTP route requires all mandatory policy predicates to return true. An escalation route must name at least one failed or unresolved predicate. Execution creates a pending-verification state; it does not create closure. Verification moves the item to closed only where the observed customer, invoice, amount, residual balance and ledger effect match the authorized expectation.

Conditional closure is reserved for situations in which the primary relationship has been executed but a non-critical follow-up remains outstanding, such as external confirmation or delayed synchronization. Its use must be explicitly authorized and time-limited. A rejected event remains available for investigation without changing the subledger. A reversed event preserves both the original and compensating actions. These controlled states make operational queues measurable: management can identify unresolved evidence, aging exceptions, verification failures and repeated rule defects without reconstructing them manually from application logs.

8. Control, Audit and Institutional Consequences

A complete GEEP population permits control testing at the level of individual automated decisions. Auditors may examine whether the configured rule operated, authority existed, exceptions were intercepted and expected state changes were verified. This does not predetermine control reliance or replace audit judgment. It creates more structured evidence for evaluating automated control operation, consistent with the growing use of automated tools and associated documentation requirements (IAASB 2020).

Where control operation cannot be supported, auditors may alter the nature, timing or extent of substantive procedures. The economic value of Rael therefore extends beyond reduced bookkeeping effort. By preserving a reconstructable evidence chain, it may affect the cost and reliability of downstream reporting, assurance and tax compliance. The same principle applies more broadly wherever public or private institutions delegate execution to AI-assisted systems: an accountable decision requires recoverable evidence, authority, action and consequence.

8.1 Relationship failures and audit assertions

The relational error taxonomy can be translated into the language of financial-statement assertions without claiming that Rael itself performs an audit. A payment assigned to the wrong customer or invoice may affect accuracy and the representation of rights and obligations. A payment applied in the wrong period affects cutoff. A posting without adequate source evidence may weaken support for occurrence or existence. Duplicate application can affect completeness and accuracy across the affected balances. An incorrect tax relationship may alter classification, valuation or disclosure depending on the reporting framework and transaction.

This mapping gives audit teams a structured route from a failed transition to the potentially affected assertion, account population and control objective. It also enables targeted testing: the auditor can select all events processed under a particular rule version, all overrides by authority class, all conditional closures, or all verification failures. The audit response remains a professional determination under the applicable assurance standards; Rael supplies a more precise and testable population from which that determination may be made.

9. Scope and Development Roadmap

Version 1.0 is a reference architecture, not an empirical validation or commercial product specification. It defines the minimum concepts, processing cycle, governance conditions and evidence payload required for a relationship-preserving implementation. Numerical thresholds, materiality rules, professional responsibilities and platform permissions must be configured within the relevant organization and jurisdiction.

The next development stage is a working accounts-receivable prototype accompanied by a reproducible benchmark. Evaluation should compare Rael with confidence-governed automation on relational accuracy, false closure, unauthorized execution, exception capture, interface-fault detection and

reconstruction quality. Later versions may extend the schema to accounts payable, inventory, payroll and cross-system operational reconciliation.

Version 1.0 design proposition

Accounting automation becomes accountable when the path from admitted evidence to relationship, rule, authority, execution, observed state and closure remains both machine-readable and professionally reconstructable.

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